§2.6 密度矩阵方法.

1. 纯态与混合态

(a) 纯态：可以用单一态矢量 $|\psi(t)\rangle$ 来描述的量子态.

一般形式：
$$|\psi(t)\rangle = c_1|\psi_1(t)\rangle + c_2|\psi_2(t)\rangle + \cdots + c_N|\psi_N(t)\rangle.$$

(b) 混合态：无法用单一态矢量来描述的量子态

态的描述方法：

$$\begin{cases} \text{参与态}：|\psi_1(t)\rangle, |\psi_2(t)\rangle, \cdots |\psi_N(t)\rangle \\ \text{出现概率}：P_1, P_2, \cdots P_N \end{cases}$$

$$\left(\sum_{i=1}^{N} P_i = 1\right) \qquad N \geq 2$$

例1：由电子枪发射出的热电子的自旋态（非极化）

$\hat{S}$ ：电子自旋算符

$$\begin{cases} \hat{S}_z|\uparrow\rangle = \frac{\hbar}{2}|\uparrow\rangle \\ \hat{S}_z|\downarrow\rangle = -\frac{\hbar}{2}|\downarrow\rangle \end{cases}$$ 基矢

设 $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$

则：$|\tilde{\psi}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \pm |\downarrow\rangle) \to \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ \pm 1\end{pmatrix}$（纯态）

$$\Rightarrow \quad \langle\tilde{\psi}|\hat{S}_z|\tilde{\psi}\rangle = \overline{S_z} = 0$$

$$\langle\tilde{\psi}|\hat{S}_y|\tilde{\psi}\rangle = \overline{S_y} = 0$$

$$\text{但} \quad \langle\tilde{\psi}|\hat{S}_x|\tilde{\psi}\rangle = \overline{S_x} = \pm\frac{\hbar}{2} \neq 0$$

$$\begin{cases} |\uparrow\rangle \to \begin{pmatrix}1\\0\end{pmatrix} \\ |\downarrow\rangle \to \begin{pmatrix}0\\1\end{pmatrix} \\ \hat{S}_x \to \frac{\hbar}{2}\begin{pmatrix}0&1\\1&0\end{pmatrix} \\ \hat{S}_y \to \frac{\hbar}{2}\begin{pmatrix}0&-i\\i&0\end{pmatrix} \\ \hat{S}_z \to \frac{\hbar}{2}\begin{pmatrix}1&0\\0&-1\end{pmatrix} \end{cases}$$

- 混合态 $\begin{cases} |\uparrow\rangle \ , \ |\downarrow\rangle & \text{（参与态）} \\ \frac{1}{2} \ , \ \frac{1}{2} & \text{（出现概率）} \end{cases}$

$$\Rightarrow \quad \overline{\hat{S}_i} = \frac{1}{2}\langle\uparrow|\hat{S}_i|\uparrow\rangle + \frac{1}{2}\langle\downarrow|\hat{S}_i|\downarrow\rangle = 0$$

$i = x, y, z.$

ok!

例2：复合系统纯态中的子系统

· 两个电子自旋系统

$\hat{S}_1$：第一个电子的自旋算符（$\hbar/2$）

$\hat{S}_2$：第二个电子的自旋算符（$\hbar/2$）

· 复合系统态矢

$$|\psi(1,2)\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow_1\downarrow_2\rangle \pm |\downarrow_1\uparrow_2\rangle\right)$$

问：第一个电子的自旋态 $|\psi_1\rangle$

怎么求？

例3：量子统计中的状态描述

2. 纯态 (Pure state) : $|\psi(t)\rangle$

(a) 纯态密度算符与密度矩阵

- 密度算符 $\hat{\rho}(t)$

$$\hat{\rho}(t) \equiv |\psi(t)\rangle\langle\psi(t)|$$

($\langle\psi(t)|\psi(t)\rangle = 1$)

投影算符 相位无关!

- 密度矩阵

A-表象: ($\hat{A}|a_n\rangle = a_n|a_n\rangle$)

$$\begin{cases} \langle a_{n'}|a_n\rangle = \delta_{n'n} \\ \sum_n |a_n\rangle\langle a_n| = \mathbb{1} \end{cases} \qquad \{|a_n\rangle\} \quad N\text{-维}$$

$$\hat{\rho}(t) = \mathbb{1}\,\hat{\rho}(t)\,\mathbb{1} = \sum_n\sum_m |a_n\rangle\langle a_n|\,\hat{\rho}(t)\,|a_m\rangle\langle a_m|$$

$$= \sum_n\sum_m |a_n\rangle\,\rho_{nm}(t)\,\langle a_m|$$

$\hat{\rho}(t) \Rightarrow \quad \rho(t) \equiv \begin{pmatrix} \vdots \\ \cdots \rho_{nm}(t) \cdots \\ \vdots \end{pmatrix}$ 密度矩阵

$N \times N$ 方矩阵

- 元素表示：$\rho_{nm}(t) = \langle a_n | \hat{\rho} | a_m \rangle = \langle a_n | \psi(t) \rangle \langle \psi(t) | a_m \rangle$

- 对角元：$\rho_{nn}(t) = |\langle a_n | \psi(t) \rangle|^2 \geq 0$

物理意义：在 $|\psi(t)\rangle$ 中测到 A 得本征值 a_n 的概率.

(b) 纯态密度算符的若干性质

① $\hat{\rho}(t)$ 是密算符 （对易于厄密矩阵）

- $\hat{\rho}(t) = |\psi(t)\rangle\langle\psi(t)|$ ，$\hat{\rho}^+(t) = |\psi(t)\rangle\langle\psi(t)| = \hat{\rho}(t)$，

$\Rightarrow$ $\hat{\rho}(t)$ 的本征值是实量. 正定性，完备性

- $\hat{\rho}^2(t) = |\psi(t)\rangle\langle\psi(t)|\psi(t)\rangle\langle\psi(t)| = |\psi(t)\rangle\langle\psi(t)| = \hat{\rho}(t)$

$\Rightarrow$ $\hat{\rho}(\hat{\rho}-1) = 0 \Rightarrow$ $\hat{\rho}(t)$ 的本征值为 0，1.

② $\mathrm{Tr}[\hat{\rho}(t)] = 1$ （与表象的选择无关）

· $\mathrm{Tr}[\hat{\rho}(t)] = \sum_n \langle a_n | \hat{\rho}(t) | a_n \rangle = \sum_n \langle a_n | \psi(t) \rangle \langle \psi(t) | a_n \rangle$

$= \sum_n \langle \psi(t) | a_n \rangle \langle a_n | \psi(t) \rangle$

$= \langle \psi(t) | \psi(t) \rangle = 1$，即 $\sum_n \rho_{nn} = 1$

· 在 $\hat{\rho}$ 的本征表象中，ρ 是对角矩阵，

但在对角元中又有一次是 1，其余皆为零.

$$\rho = \begin{bmatrix} \ddots & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \ddots \end{bmatrix}$$

③ $\mathrm{Tr}[\hat{\rho}^2(t)] = 1$，反之 $\hat{\rho}(t) = \hat{\rho}(t)$

$\Rightarrow \mathrm{Tr}[\hat{\rho}^2(t)] = \mathrm{Tr}[\hat{\rho}(t)] = 1$

④ 正定性：对任意态，$\hat{\rho}(t)$ 的平均值 ≥ 0

· $\overline{\hat{\rho}(t)} = \langle \varphi(t) | \hat{\rho}(t) | \varphi(t) \rangle = \langle \varphi(t) | \psi(t) \rangle \langle \psi(t) | \varphi(t) \rangle$

$= | \langle \varphi(t) | \psi(t) \rangle |^2 \geq 0$

(C) 力学量的平均值用 $\hat{\rho}(t)$ 表示

- $$\overline{\hat{B}(t)} \equiv \langle \psi(t) | \hat{B} | \psi(t) \rangle$$

$$= \sum_n \sum_m \langle \psi(t) | a_n \rangle \langle a_n | \hat{B} | a_m \rangle \langle a_m | \psi(t) \rangle$$

$$= \sum_n \sum_m \langle a_m | \psi(t) \rangle \langle \psi(t) | a_n \rangle \langle a_n | \hat{B} | a_m \rangle$$

$$= \sum_n \sum_m \langle a_m | \hat{\rho}(t) | a_n \rangle \langle a_n | \hat{B} | a_m \rangle$$

$$= \sum_m \langle a_m | \hat{\rho}(t) \hat{B} | a_m \rangle$$

$$= Tr[\hat{\rho}(t)\hat{B}] = Tr[\hat{B}\hat{\rho}(t)]$$

即

$$\boxed{\langle \psi(t) | \hat{B} | \psi(t) \rangle = Tr[\hat{\rho}(t)\hat{B}]}$$

与表象无关！

例：设一电子处于自旋态（基矢：$|\uparrow\rangle, |\downarrow\rangle$）

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left[e^{i\phi}|\uparrow\rangle + |\downarrow\rangle \right] 中.$$

求 $\hat{\rho}$、ρ 以及平均值 $\overline{\hat{S}_x}$.

(1) $\hat{\rho} = |\psi\rangle\langle\psi|$

$$= \left\{ \frac{1}{\sqrt{2}}\left(e^{i\phi}|\uparrow\rangle + |\downarrow\rangle \right) \right\} \left\{ \frac{1}{\sqrt{2}}\left(e^{-i\phi}\langle\uparrow| + \langle\downarrow| \right) \right\}$$

$$= \frac{1}{2}\Big[|\uparrow\rangle\langle\uparrow| + e^{i\phi}|\uparrow\rangle\langle\downarrow| + e^{-i\phi}|\downarrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow| \Big]$$

(2.-1)

$$\rho \equiv \begin{array}{c} \\ \langle\uparrow| \\ \langle\downarrow| \end{array} \begin{array}{c} \begin{array}{cc} |\uparrow\rangle & |\downarrow\rangle \end{array} \\ \begin{pmatrix} \langle\uparrow|\hat{\rho}|\uparrow\rangle & \langle\uparrow|\hat{\rho}|\downarrow\rangle \\ \langle\downarrow|\hat{\rho}|\uparrow\rangle & \langle\downarrow|\hat{\rho}|\downarrow\rangle \end{pmatrix} \end{array}$$

$$\mathcal{O}_{\mathbb{P}^1}(-1) \cong \mathcal{O}_{\mathbb{P}^1}(-1)$$

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- 基矢： $|\uparrow\rangle$ 、 $|\downarrow\rangle$.

- 完备性： $|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow| = \mathbb{1}$

- $$\hat{S}_x = \mathbb{1}\,\hat{S}_x\,\mathbb{1}$$

$$= (|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|)\,\hat{S}_x\,(|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|)$$

$$= |\uparrow\rangle\langle\uparrow|\hat{S}_x|\uparrow\rangle\langle\uparrow| + |\uparrow\rangle\langle\uparrow|\hat{S}_x|\downarrow\rangle\langle\downarrow|$$

$$+ |\downarrow\rangle\langle\downarrow|\hat{S}_x|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|\hat{S}_x|\downarrow\rangle\langle\downarrow| .$$

但， $S_x = \begin{bmatrix} \langle\uparrow|\hat{S}_x|\uparrow\rangle & \langle\uparrow|\hat{S}_x|\downarrow\rangle \\ \langle\downarrow|\hat{S}_x|\uparrow\rangle & \langle\downarrow|\hat{S}_x|\downarrow\rangle \end{bmatrix} = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\Rightarrow \hat{S}_x = \frac{\hbar}{2}\left[|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow|\right].$$

类似， $S_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ 、 $S_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\Downarrow \qquad \Downarrow$$

$$\hat{S}_y = -\frac{i\hbar}{2}\left[|\uparrow\rangle\langle\downarrow| - |\downarrow\rangle\langle\uparrow|\right] \quad , \quad \hat{S}_z = \frac{\hbar}{2}\left[|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|\right]$$

(3-1) $\overline{\hat{S}_x} = Tr[\hat{\rho}\hat{S}_x]$

$$\hat{\rho}\hat{S}_x = \left\{ \frac{1}{2}\left[|\uparrow\rangle\langle\uparrow| + e^{i\phi}|\uparrow\rangle\langle\downarrow| + e^{-i\phi}|\downarrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow| \right] \right\} \hat{\rho}$$

$$\times \underbrace{\left\{ \frac{\hbar}{2}\left(|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow| \right) \right\}}_{\hat{S}_x \text{ 가} \text{ 행렬 표시}}$$

$$= \frac{\hbar}{4}\left[|\uparrow\rangle\langle\downarrow| + e^{i\phi}|\uparrow\rangle\langle\uparrow| + e^{-i\phi}|\downarrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow| \right]$$

$$\Rightarrow \quad Tr[\hat{\rho}\hat{S}_x] = \sum_{\sigma=\uparrow,\downarrow} \langle\sigma|\hat{\rho}\hat{S}_x|\sigma\rangle$$

$$= \langle\uparrow|\hat{\rho}\hat{S}_x|\uparrow\rangle + \langle\downarrow|\hat{\rho}\hat{S}_x|\downarrow\rangle = \frac{\hbar}{4}\left[e^{i\phi} + e^{-i\phi}\right] = \frac{\hbar}{2}\cos\phi .$$

(3-2) $$\mathrm{Tr}[\hat{\rho}\hat{S}_x] = \sum_{\sigma=\uparrow,\downarrow} \langle\sigma|\hat{\rho}\hat{S}_x|\sigma\rangle$$

$$= \sum_{\sigma}\sum_{\sigma'} \langle\sigma|\hat{\rho}|\sigma'\rangle\langle\sigma'|\hat{S}_x|\sigma\rangle$$

$$= \sum_{\sigma}\sum_{\sigma'} \rho_{\sigma\sigma'} S_{x,\sigma'\sigma}$$

等于两个矩阵的乘积 ρS_x 的对角元之和.

$$\rho S_x = \frac{1}{2}\begin{pmatrix} 1 & e^{i\phi} \\ e^{-i\phi} & 1 \end{pmatrix}\frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \frac{\hbar}{4}\begin{pmatrix} e^{i\phi} & 1 \\ 1 & e^{-i\phi} \end{pmatrix}$$

$$\Rightarrow \quad \mathrm{Tr}[\hat{\rho}\hat{S}] = \frac{\hbar}{4}\left(e^{i\phi} + e^{-i\phi}\right)$$

$$= \frac{\hbar}{2}\cos\phi .$$

(d) $\hat{\rho}(t)$ の時間変化.

$\hat{\rho}(t) = |\psi(t)\rangle\langle\psi(t)|$, $i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle$ を満たす.

$$i\hbar\frac{d}{dt}\hat{\rho}(t) = \left(i\hbar\frac{d}{dt}|\psi(t)\rangle\right)\langle\psi(t)|$$

$$+ |\psi(t)\rangle\left(i\hbar\frac{d}{dt}\langle\psi(t)|\right)$$

$$= \hat{H}(t)|\psi(t)\rangle\langle\psi(t)|$$

$$- |\psi(t)\rangle\langle\psi(t)|\hat{H}(t)$$

$$= \hat{H}(t)\hat{\rho}(t) - \hat{\rho}(t)\hat{H}(t)$$

$$= [\hat{H}(t), \hat{\rho}(t)].$$

よって

$$\boxed{i\hbar\frac{d}{dt}\hat{\rho}(t) = [\hat{H}(t), \hat{\rho}(t)]}$$

Neumann 方程式.

初始条件： $\hat{\rho}(t_0) = |\psi(t_0)\rangle\langle\psi(t_0)|$.

. $|\psi(t)\rangle = \hat{U}(t\,t_0)|\psi(t_0)\rangle$, $\langle\psi(t)| = \langle\psi(t_0)|\hat{U}^{+}(t\,t_0)$

$\Rightarrow \hat{\rho}(t) = \hat{U}(t\,t_0)|\psi(t_0)\rangle\langle\psi(t_0)|\hat{U}^{+}(t\,t_0)$

$= \hat{U}(t\,t_0)\hat{\rho}(t_0)\hat{U}^{+}(t\,t_0)$.

即

$$\hat{\rho}(t) = \hat{U}(t\,t_0)\hat{\rho}(t_0)\hat{U}^{+}(t\,t_0)$$

3. 混合态 (Mixed state)

态的描述方式:

$$\begin{cases} |\psi_1\rangle, |\psi_2\rangle, \cdots |\psi_s\rangle & \text{参与态} \\ p_1, p_2, \cdots p_s \quad (s \geq 2) & \text{权重.} \end{cases}$$

$$\left(0 < p_i < 1, \quad \sum_{i=1}^{s} p_i = 1 \right)$$

· 参与态数目 s 可大于希尔伯特空间维数 N.

$$\begin{cases} |\uparrow\rangle, & |\nearrow\rangle, & |\leftarrow\rangle, & |\downarrow\rangle \\ 0.3 & 0.2 & 0.1 & 0.4 \end{cases} \quad \left(\text{等效} \; |\uparrow\rangle, |\downarrow\rangle \right)$$

·
$$\begin{cases} \langle\psi_i|\psi_i\rangle = 1 \\ \langle\psi_j|\psi_i\rangle \neq 0 \quad (i \neq j) \end{cases}$$ 可不正交!

(a) 混合态的密度算符与密度矩阵

- 密度算符 $\hat{\rho}_M(t)$

$$\hat{\rho}_M(t) \equiv \sum_{i=1}^{N} p_i \, |\psi_i(t)\rangle\langle\psi_i(t)| \qquad (N \geq 2)$$

- 密度矩阵 （A-表象）

$$\hat{\rho}_M(t) = \sum_n^N \sum_m^N |a_n\rangle \rho_{M,nm}(t) \langle a_m| \,.$$

其中，$\rho_{M,nm}(t) = \langle a_n | \hat{\rho}_M(t) | a_m \rangle$.

$$= \sum_{i=1} p_i \langle a_n|\psi_i(t)\rangle\langle\psi_i(t)|a_m\rangle$$

（矩阵元）

密度矩阵：

$$\begin{bmatrix} \cdots & \vdots & \cdots \\ \cdots & \rho_{M,nm}^{(t)} & \cdots \\ \cdots & \vdots & \cdots \end{bmatrix} \equiv \rho_M^{(t)}$$

$N \times N$ 矩阵.

- 对角元：$\rho_{M,nn}(t) = \sum_i p_i \left| \langle a_n | \psi_i(t) \rangle \right|^2$

$$\left(= Tr\left[\hat{\rho}_M |a_n\rangle\langle a_n| \right] \right)$$

物理意义：在该混合态中测量力学量 A，得值为 a_n 的概率.

- 非对角元：$\rho_{M,nm}^{(t)} = \sum_i p_i \langle a_n | \psi_i(t) \rangle \langle \psi_i(t) | a_m \rangle$

$$\left(= Tr\left[\hat{\rho}_M |a_m\rangle\langle a_n| \right] \right).$$

物理意义：

体现在该混合态中，$|a_n\rangle$ 态与 $|a_m\rangle$ 态的干涉效应.

(b) $\hat{\rho}_M$ 的若干性质

① $\hat{\rho}_M^\dagger = \hat{\rho}_M$ （厄密算符）

⇒ 本征值为实数、正交性、完备性。

② $\hat{\rho}_M^2 \neq \hat{\rho}_M(t)$.

. $\hat{\rho}_M^2 = \left[\sum_k p_k |\psi_k\rangle\langle\psi_k|\right]\left[\sum_k p_k |\psi_k\rangle\langle\psi_k|\right]$

$= \sum_k \sum_k p_k p_k |\psi_k\rangle\langle\psi_k|\psi_k\rangle\langle\psi_k| \neq \hat{\rho}_M$

③ $Tr[\hat{\rho}_M] = 1$

. $Tr[\hat{\rho}_M] = \sum_n \langle a_n|\hat{\rho}_M|a_n\rangle$

$= \sum_n \langle a_n| \sum_i p_i |\psi_i\rangle\langle\psi_i| a_n\rangle$

$= \sum_n \sum_i p_i \langle\psi_i|a_n\rangle\langle a_n|\psi_i\rangle$

$= \sum_i p_i \langle\psi_i|\psi_i\rangle$

$= \sum_i p_i = 1$ 。

(4) $\mathrm{Tr}[\hat{\rho}_M^2(t)] < 1$

证明: $\hat{\rho}_M^2 = \left(\sum_a P_a |\psi_a\rangle\langle\psi_a|\right)\left(\sum_k P_k |\psi_k\rangle\langle\psi_k|\right)$

$$\mathrm{Tr}[\hat{\rho}_M^2] = \sum_n \sum_a \sum_k P_a P_k \langle a_n|\psi_a\rangle\langle\psi_a|\psi_k\rangle\langle\psi_k|a_n\rangle$$

$$= \sum_n \sum_a \sum_k P_a P_k \langle\psi_k|a_n\rangle\langle a_n|\psi_a\rangle\langle\psi_a|\psi_k\rangle$$

$$= \sum_a \sum_k P_a P_k \langle\psi_k|\psi_a\rangle\langle\psi_a|\psi_k\rangle$$

$$= \sum_a P_a \left[\sum_k P_k \langle\psi_k|\psi_a\rangle\langle\psi_a|\psi_k\rangle\right] \equiv \sum_a P_a F_a$$

其中 $F_a \equiv \sum_k P_k \langle\psi_k|\psi_a\rangle\langle\psi_a|\psi_k\rangle$

$$= P_a \langle\psi_a|\psi_a\rangle\langle\psi_a|\psi_a\rangle$$

$$+ \sum_{k\neq a} P_k \langle\psi_k|\psi_a\rangle\langle\psi_a|\psi_k\rangle$$

$$= P_\alpha + \sum_{k\neq\alpha} P_k \langle\psi_k|\psi_\alpha\rangle\langle\psi_\alpha|\psi_k\rangle$$

$$= P_\alpha + \sum_{k\neq\alpha} P_k |\langle\psi_k|\psi_\alpha\rangle|^2$$

但根据 Schwartz 不等式，有

$$|\langle\psi_m|\psi_\alpha\rangle|^2 < \left(\langle\psi_m|\psi_k\rangle\langle\psi_\alpha|\psi_\alpha\rangle\right) = 1$$

$$\Rightarrow F_\alpha < P_\alpha + \sum_{k\neq\alpha} P_k \cdot 1 = \sum_k P_k = 1$$

$$\Rightarrow \mathrm{Tr}\left[\hat{\rho}_M^2\right] = \sum_\alpha P_\alpha F_\alpha < \sum_\alpha P_\alpha = 1$$

$$\Rightarrow \mathrm{Tr}\left[\hat{\rho}_M^2\right] < 1$$

证毕.

(4) 正定性 ： 对于任意量子态 $|\varphi(t)\rangle$，有

$$\overline{\hat{\rho}_M} = \langle \varphi(t) | \hat{\rho}_M(t) | \varphi(t) \rangle$$

$$= \langle \varphi(t) | \sum_i p_i |\psi_i(t)\rangle\langle\psi_i(t)| \varphi(t) \rangle$$

$$= \sum_i p_i \ |\langle \varphi(t) | \psi_i(t) \rangle|^2 \geq 0 \ .$$

(5) $\hat{\rho}_M$ 的本征值的取值范围

. $\hat{\rho}_M |\rho_n\rangle = \rho_n |\rho_n\rangle$. 则有

$$0 \leq \rho_n < 1 \ .$$

证明：

本征方程为：$\hat{\rho}_M |\rho_n\rangle = \rho_n |\rho_n\rangle$

(i) $\langle \rho_n | \hat{\rho}_M | \rho_n \rangle = \underline{\rho_n \geq 0}$ （正定性）

(ii) ρ_n 为实数

$$\underline{Tr[\hat{\rho}_M^2]} = \sum_n \langle \rho_n | \hat{\rho}_M^2 | \rho_n \rangle = \sum_n \rho_n^2 \underline{< 1}$$

$$\Rightarrow \quad \rho_n^2 < 1 \quad \Rightarrow \quad \rho_n < 1 \quad \Rightarrow ?$$

$$\underline{0 \leq \rho_n < 1}$$

(C) 物理量的平均值

$$\overline{\hat{B}} \equiv \sum_i p_i \langle \psi_i | \hat{B} | \psi_i \rangle$$

$$= \sum_n \sum_i p_i \langle \psi_i | \hat{B} | a_n \rangle \langle a_n | \psi_i \rangle$$

$$= \sum_n \sum_i p_i \langle a_n | \psi_i \rangle \langle \psi_i | \hat{B} | a_n \rangle$$

$$= \sum_n \langle a_n | \left[\sum_i p_i | \psi_i \rangle \langle \psi_i | \right] \hat{B} | a_n \rangle$$

$$= \sum_n \langle a_n | \hat{\rho}_M \hat{B} | a_n \rangle = \mathrm{Tr}\left(\hat{\rho}_M \hat{B} \right)$$

Exp.

$$\overline{\hat{B}} = \mathrm{Tr}\left[\hat{\rho}_M \hat{B} \right]$$

(d) $\hat{\rho}_M(t)$ 的时间演化 $\left(\frac{\partial}{\partial t} p_i = 0\right)$

- $i\hbar \frac{d}{dt} \hat{\rho}_M(t) = \left[\hat{H}(t), \hat{\rho}_M(t) \right]$, Neumann 方程

- $\hat{\rho}_M(t) = \hat{U}(t, t_0)\, \hat{\rho}_M(t_0)\, \hat{U}^{+}(t, t_0)$.

统一描述

4 纯态与混合态的特性.

例：电子自旋态 （基矢：$|\uparrow\rangle$, $|\downarrow\rangle$）

(a) 纯态： $|\psi\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle$

（概率幅描述）

- 密度算符： $\hat{\rho} = |\psi\rangle\langle\psi|$

$$= \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow| + \frac{1}{2}|\uparrow\rangle\langle\downarrow| + \frac{1}{2}|\downarrow\rangle\langle\uparrow|$$

干涉项

- 密度矩阵：

$$\rho = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{cases} Tr(\hat{\rho}) = 1 \\ Tr(\hat{\rho}^2) = 1 \end{cases} , \begin{cases} \hat{\rho}^2 = \hat{\rho} \ , \ (\rho^2 = \rho) \\ \rho \text{ 的本征值} : 1, 0 \end{cases}$$

(b) 混合态、 $\begin{cases} \text{多与态} & |\uparrow\rangle \quad , \quad |\downarrow\rangle \\ p & \frac{1}{2} \quad , \quad \frac{1}{2} \end{cases}$ (概率描述)

密度算符 : $\hat{\rho}_M = \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow|$

密度矩阵 :

$$\rho_M = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$ (对角矩阵)

$$\begin{cases} Tr[\hat{\rho}_M] = 1 \\ Tr[\hat{\rho}_M^2] = \frac{1}{2} < 1 \end{cases}, \quad \begin{cases} \rho_M^2 = \frac{1}{2}\rho_M \neq \rho_M \\ \rho_M \text{的本征值: } \frac{1}{2}, \frac{1}{2}. \end{cases}$$

4 物理特性 （基矢：$|\uparrow\rangle$, $|\downarrow\rangle$.）

· 计算 $\hat{S}_x$ 的平均值.

· $\hat{S}_x = \frac{\hbar}{2}\left(|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow| \right)$

· $\langle\uparrow|\hat{S}_x|\uparrow\rangle = 0$, $\langle\downarrow|\hat{S}_x|\downarrow\rangle = 0$

(a) 纯态平均 （$|\psi\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle$）

$$\overline{\hat{S}_x} = \langle\psi|\hat{S}_x|\psi\rangle$$

$$= \frac{1}{2}\langle\uparrow|\hat{S}_x|\uparrow\rangle + \frac{1}{2}\langle\downarrow|\hat{S}_x|\downarrow\rangle$$

$$+ \frac{1}{2}\langle\uparrow|\hat{S}_x|\downarrow\rangle + \frac{1}{2}\langle\downarrow|\hat{S}_x|\uparrow\rangle = \frac{\hbar}{2}.$$ (相干迭加!)

(b) 混合态平均

{ 参与态: $|\uparrow\rangle$. $|\downarrow\rangle$
  P $\frac{1}{2}$. $\frac{1}{2}$

$$\overline{S_x} = \sum_i p_i \langle\psi_i|\hat{S}_x|\psi_i\rangle$$

$$= \frac{1}{2}\langle\uparrow|\hat{S}_x|\uparrow\rangle + \frac{1}{2}\langle\downarrow|\hat{S}_x|\downarrow\rangle = 0$$

（非相干迭加!）

* $\hat{\rho}_m$ 的参与态组合的非唯一性

(可给予不同的物理解释)

例1: $\hat{\rho}_m = \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow|$

$$= \frac{1}{2}|\rightarrow\rangle\langle\rightarrow| + \frac{1}{2}|\leftarrow\rangle\langle\leftarrow|$$

$$= \frac{1}{2}|\otimes\rangle\langle\otimes| + \frac{1}{2}|\odot\rangle\langle\odot|$$

$$= \frac{1}{2}|x\rangle\langle x| + \frac{1}{2}|y\rangle\langle y|$$

$$\vdots$$

其中. $\begin{cases} |\uparrow\rangle_x = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) \\ |\downarrow\rangle_x = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle), \end{cases}$

$\begin{cases} |\uparrow\rangle_y = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle) \\ |\downarrow\rangle_y = \frac{1}{\sqrt{2}}(|\uparrow\rangle - i|\downarrow\rangle), \end{cases}$ (正交)

$\begin{cases} |x\rangle = \frac{1}{\sqrt{3}}|\uparrow\rangle - \sqrt{\frac{2}{3}}|\downarrow\rangle \\ |y\rangle = \sqrt{\frac{2}{3}}|\uparrow\rangle + \frac{1}{\sqrt{3}}|\downarrow\rangle. \end{cases}$

$$\vdots$$

$|\psi|^2$:

$$\hat{\rho}_M = \frac{1}{3}|\uparrow\rangle\langle\uparrow| + \frac{2}{3}|\downarrow\rangle\langle\downarrow|$$

$$= \frac{1}{2}|\psi_1\rangle\langle\psi_1| + \frac{1}{2}|\psi_2\rangle\langle\psi_2|$$

$$= \frac{4}{9}|\psi_3\rangle\langle\psi_3| + \frac{5}{9}|\psi_4\rangle\langle\psi_4|$$

$$\vdots$$

其中 $\begin{cases} |\psi_1\rangle \equiv \frac{1}{\sqrt{3}}\left(|\uparrow\rangle + \sqrt{2}|\downarrow\rangle\right) \\ |\psi_2\rangle \equiv \frac{1}{\sqrt{3}}\left(|\uparrow\rangle - \sqrt{2}|\downarrow\rangle\right), \end{cases}$ $(\langle\psi_1|\psi_2\rangle \neq 0)$

$\begin{cases} |\psi_3\rangle \equiv \frac{1}{\sqrt{2}}\left(|\uparrow\rangle + |\downarrow\rangle\right) \\ |\psi_4\rangle \equiv \frac{1}{\sqrt{5}}\left(|\uparrow\rangle - 2|\downarrow\rangle\right). \end{cases}$ $(\langle\psi_3|\psi_4\rangle \neq 0)$

(相加的话2)

混合态的密度矩阵不一定总是对角的！

例1：$\hat{\rho}_n = \frac{1}{3}|\uparrow\rangle\langle\uparrow| + \frac{2}{3}|\downarrow\rangle\langle\downarrow|$

(i) S_z-表象（Pauli表象）：

基矢：$|\uparrow\rangle$，$|\downarrow\rangle$

$$\rho_M = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{2}{3} \end{pmatrix}$$ （参与态为基矢）

(ii) S_x-表象：（参与态不是基矢）

基矢：$|\uparrow\rangle_x$，$|\downarrow\rangle_x$

$$|\uparrow\rangle_x \equiv \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle),$$

$$|\downarrow\rangle_x \equiv \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle),$$

$$\left(\hat{S}_x|\uparrow\rangle_x = \frac{\hbar}{2}|\uparrow\rangle_x, \quad \hat{S}_x|\downarrow\rangle_x = -\frac{\hbar}{2}|\downarrow\rangle_x\right).$$

$$\Rightarrow \quad \rho_m = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{2} \end{pmatrix} \qquad (\text{행렬표시})$$

$$= \frac{1}{2}|\uparrow\rangle\langle\uparrow| - \frac{1}{6}|\uparrow\rangle\langle\downarrow| + \frac{1}{2}|\downarrow\rangle\langle\uparrow| - \frac{1}{6}|\downarrow\rangle\langle\downarrow|$$

$$= \left\{\frac{1}{2}(|\uparrow\rangle - |\downarrow\rangle)\right\}\left\{\frac{1}{2}(\langle\uparrow| - \langle\downarrow|)\right\} + \frac{2}{3}\left\{\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)\right\}\left\{\frac{1}{\sqrt{2}}(\langle\uparrow| + \langle\downarrow|)\right\}$$

$$= \frac{1}{3}\left\{\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)\right\}\left\{\frac{1}{\sqrt{2}}(\langle\uparrow| + \langle\downarrow|)\right\}$$

$$\Rightarrow \quad \hat{\rho}_m = \frac{1}{3}|\uparrow\rangle\langle\uparrow| + \frac{2}{3}|\downarrow\rangle\langle\downarrow|$$

$$\Leftarrow \quad |\uparrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) \quad ,$$

$$|\downarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle) \quad .$$

例2：$\hat{\rho}_M = \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{4}|\downarrow\rangle\langle\downarrow| + \frac{1}{4}|\rightarrow\rangle_x\langle\rightarrow|$

其中 $|\rightarrow\rangle_x = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$，$(\hat{S}_x|\rightarrow\rangle_x = \frac{\hbar}{2}|\rightarrow\rangle_x)$

写为：

$$\rho_M = \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{4}|\downarrow\rangle\langle\downarrow| + \frac{1}{8}(|\uparrow\rangle\langle\uparrow| + |\rightarrow\rangle\langle\uparrow| + |\uparrow\rangle\langle\rightarrow| + |\rightarrow\rangle\langle\rightarrow|)$$

$$= \frac{5}{8}|\uparrow\rangle\langle\uparrow| + \frac{1}{8}|\rightarrow\rangle\langle\uparrow| + \frac{1}{8}|\uparrow\rangle\langle\rightarrow| + \frac{3}{8}|\rightarrow\rangle\langle\rightarrow|$$

S_z 表象中，有

$$\rho_M = \begin{pmatrix} \frac{5}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{3}{8} \end{pmatrix}$$